

Convergence of different ADC(n) variants through $n = 15$

These notes builds on the Dyson vs non-Dyson controversy, and assesses how and if the different ways, known to the literature, to generate a systematic ADC converge to the FCI limit.

I. THE λ -SCALED PROBLEM

Sector I are the $(N - 1)$ configurations (2h1p, ...), sector II the $(N + 1)$ ones (2p1h, ...). All results are properties of the hamiltonian

$$\hat{H}(\lambda) = \hat{F} + \lambda(\hat{H} - \hat{F}), \quad (1)$$

where

$$\hat{F} = \sum_p \epsilon_p \hat{a}_p^\dagger \hat{a}_p \quad (2)$$

is the canonical HF Fock operator, built as dense matrices in the determinant spaces $\mathcal{H}_N$ and $\mathcal{H}_{N\pm 1}$ of the systems in Table I. FCI gives the exact $\Sigma(\omega; \lambda)$ at any λ ; its Taylor coefficients follow from complex λ on a circle of radius $r = 0.3$ with $N_\lambda = 32$ points,¹

$$\Sigma^{(m)}(\omega) = \frac{1}{N_\lambda r^m} \sum_{j=0}^{N_\lambda-1} \Sigma(\omega; r e^{2\pi i j / N_\lambda}) e^{-2\pi i m j / N_\lambda}. \quad (3)$$

The only perturbative input to ADC is the Rayleigh-Schrödinger (RS) series of the ground state and its energy,

$$|\Psi_0(\lambda)\rangle = \sum_{m \geq 0} \lambda^m |\Psi_0^{(m)}\rangle, \quad (4)$$

$$E_0(\lambda) = \sum_{m \geq 0} \lambda^m E_0^{(m)}. \quad (5)$$

The zeroth-order state is the HF determinant $|\Phi_0\rangle$, which is an eigenstate of $\hat{F}$,

$$\hat{F} |\Phi_0\rangle = E_0^{(0)} |\Phi_0\rangle, \quad (6)$$

with $E_0^{(0)}$ the sum of the occupied orbital energies. We use intermediate normalization, *i.e.* every correction is orthogonal to the HF determinant,

$$\langle \Phi_0 | \Psi_0^{(m)} \rangle = \delta_{m0}. \quad (7)$$

Inserting the two series into the Schrödinger equation of $\hat{H}(\lambda)$ and collecting the terms of order λ^m gives the recursion

$$|\Psi_0^{(m)}\rangle = \hat{R}_0 \left[(\hat{H} - \hat{F}) |\Psi_0^{(m-1)}\rangle - \sum_{l=1}^m E_0^{(l)} |\Psi_0^{(m-l)}\rangle \right]. \quad (8)$$

Here $\hat{R}_0$ is the reduced resolvent of $\hat{F}$,

$$\hat{R}_0 = \hat{Q} (E_0^{(0)} - \hat{F})^{-1} \hat{Q}, \quad (9)$$

where

$$\hat{Q} = 1 - |\Phi_0\rangle \langle \Phi_0| \quad (10)$$

projects out the HF determinant. Because $\hat{F}$ is diagonal in the determinant basis, applying $\hat{R}_0$ amounts to dividing each determinant coefficient by the difference between $E_0^{(0)}$ and the orbital-energy sum of that determinant. The projector keeps the corrections orthogonal to $|\Phi_0\rangle$ and removes the one vanishing denominator. The energy corrections follow from projecting the order- l equation onto $|\Phi_0\rangle$,

$$E_0^{(l)} = \langle \Phi_0 | \hat{H} - \hat{F} | \Psi_0^{(l-1)} \rangle. \quad (11)$$

Since the matrix elements of Sec. II need the normalized state, every series is finally divided by the series of $\langle \Psi_0 | \Psi_0 \rangle$.

TABLE I. Systems (STO-3G). n_{occ} , n_{vir} : occupied and virtual spin orbitals; k_{max} : last non-empty class; $n_c = 2k_{\text{max}}$: first order with the complete manifold; n_{op} : size of the complete manifold.

system	geometry (Å)	n_{occ}	n_{vir}	k_{max}	n_c	n_{op}
H ₄ chain	spacing 1.0	4	4	3	6	112
H ₄ chain	spacing 1.6	4	4	3	6	112
LiH	1.6	4	8	4	8	1012
H ₆ chain	spacing 1.0	6	6	5	10	1584

II. INTERMEDIATE-STATE BLOCKS

The excitation operators of the two sectors are the ionizing strings

$$\{\hat{r}_v^-\} = \{\hat{a}_b^\dagger \hat{a}_j \hat{a}_i, \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_k \hat{a}_j \hat{a}_i, \dots\} \quad (12)$$

of sector I and the attaching strings

$$\{\hat{r}_v^+\} = \{\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j, \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \hat{a}_k \hat{a}_j, \dots\} \quad (13)$$

of sector II. The ADC supermatrix is labelled by the single-character manifold

$$\{\hat{X}_J\} = \{\hat{a}_p^\dagger\} \cup \{\hat{r}_v^+\} \cup \{\hat{r}_v^-\}, \quad (14)$$

in which every operator adds one electron. It is ordered into classes k : $\hat{a}_p^\dagger$ is class 0; class k holds the $(k+1)$ pkh strings of sector II and the adjoints of the $(k+1)$ hpk strings of sector I. Each $\hat{X}_J$ becomes the vector pair $(\hat{X}_J \Psi_0, \hat{X}_J^\dagger \Psi_0)$ in $\mathcal{H}_{N+1} \oplus \mathcal{H}_{N-1}$, so that

$$\mathbb{S}_{IJ} = \langle \hat{X}_I \Psi_0 | \hat{X}_J \Psi_0 \rangle + \langle \hat{X}_J^\dagger \Psi_0 | \hat{X}_I^\dagger \Psi_0 \rangle \quad (15)$$

is the metric $\langle \{\hat{X}_I^\dagger, \hat{X}_J\} \rangle$ and

$$\mathbb{M}_{IJ} = \langle \hat{X}_I \Psi_0 | \hat{H} - E_0 | \hat{X}_J \Psi_0 \rangle - \langle \hat{X}_J^\dagger \Psi_0 | \hat{H} - E_0 | \hat{X}_I^\dagger \Psi_0 \rangle \quad (16)$$

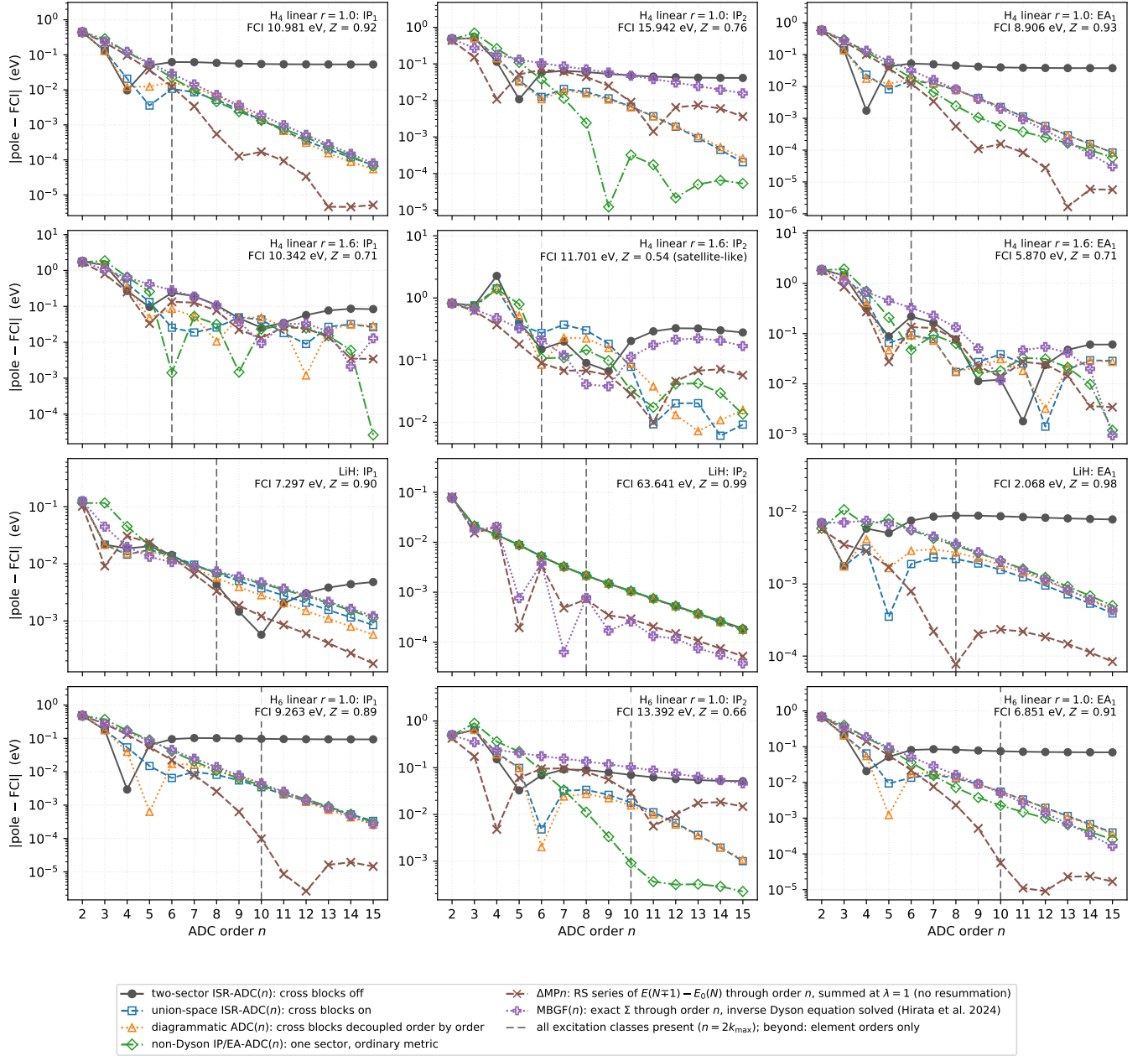


FIG. 1. Principal ADC(n) poles against FCI at $\lambda = 1$. Dashed line: $n = 2k_{\max}$, beyond which the manifold is complete (for LiH the non-Dyson IP sector is complete at $n = 6$).

is the superoperator $\langle \{\hat{X}_I^\dagger, [\hat{H}, \hat{X}_J]\} \rangle$. Inserting the RS series gives $\mathbb{S}^{(m)}$ and $\mathbb{M}^{(m)}$ exactly. The intermediate states²⁻⁴ follow from Gram-Schmidt of class k against all lower classes and Löwdin orthonormalization within it, both done on the power series. The orthonormal supermatrix is

$$\tilde{\mathbb{M}} = \begin{pmatrix} \epsilon + \Sigma(\infty) & \mathbf{U}^I & \mathbf{U}^{II} \\ \mathbf{U}^{I\dagger} & -\mathbf{K}^I - \mathbf{C}^I & \mathbf{X} \\ \mathbf{U}^{II\dagger} & \mathbf{X}^\dagger & \mathbf{K}^{II} + \mathbf{C}^{II} \end{pmatrix}, \quad (17)$$

where $\mathbf{K}$ is the diagonal zeroth order and $\mathbf{X}$ the cross block between the sectors (its metric counterpart vanishes identically).

The inner projection⁵ $\mathbf{G} = [\mathbb{S}(\omega\mathbb{S} - \tilde{\mathbb{M}})^{-1}\mathbb{S}]_{00}$ then has the Dyson form with

$$\Sigma(\omega) = \Sigma(\infty) + \mathbf{U}(\omega - \tilde{\mathbb{M}}_{\text{cc}})^{-1}\mathbf{U}^\dagger, \quad (18)$$

$\tilde{\mathbb{M}}_{\text{cc}}$ being all classes $k \geq 1$. Over the complete manifold this reproduces FCI to 10^{-13} . Folded order by order, the dynamic part $\Sigma_{\text{dyn}} = \Sigma - \Sigma(\infty)$ is

$$\Sigma_{\text{dyn}}^{(m)}(\omega) = \sum_{a+b+c=m} \mathbf{U}^{(a)} \mathbf{R}^{(b)}(\omega) \mathbf{U}^{(c)\dagger}, \quad (19)$$

where $\mathbf{R}^{(b)}$ is the order- b coefficient of $(\omega - \tilde{\mathbb{M}}_{\text{cc}})^{-1}$. This is the exact Taylor reference; it agrees with the contour coefficients

to 5×10^{-13} . $\text{ADC}(n)^{6,7}$ keeps classes $k \leq \lfloor n/2 \rfloor$ and block (k, l) through order $n - k - l$,

$$[\tilde{\mathbb{M}}_{\text{ADC}(n)}]_{kl} = \sum_{m=0}^{n-k-l} \tilde{\mathbb{M}}_{kl}^{(m)}. \quad (20)$$

Its eigenvalues at $\lambda = 1$ are the poles; principal poles have orbital weight $Z > 0.3$. For $n \geq n_c$ (Table I) the manifold is complete and higher n only adds orders to its matrix elements.

III. CONSTRUCTIONS

All Dyson variants share $\Sigma(\infty)$, $\mathbf{U}$ and the same-sector blocks exactly; they differ in $\mathbf{X}$.

TABLE II. The constructions of Fig. 1. Limit: complete manifold, all orders.

construction	$\mathbf{X}$	exact through	limit
two-sector $\text{ADC}(n)$	dropped	λ^3	not FCI
union $\text{ADC}(n)$	Eq. (20)	λ^n	FCI
diagrammatic $\text{ADC}(n)$	rotated	λ^n	FCI
non-Dyson $\text{ADC}(n)$	none	λ^n	FCI
$\Delta\text{MP}n$	–	λ^n	RS series
$\text{MBGF}(n)$	–	Σ to λ^n	Dyson root

A. Union-space ISR- $\text{ADC}(n)$

Union-space ISR- $\text{ADC}(n)$ keeps $\mathbf{X}$ under the rule of Eq. (20). The block between class k of sector I and class l of sector II starts at order

$$L_{kl} = \left\lceil \frac{k+l+1}{2} \right\rceil, \quad (21)$$

so under the rule the block $\mathbf{X}_{11}$ enters at $n = 4$, $\mathbf{X}_{12}$ at $n = 5$, $\mathbf{X}_{22}$ and $\mathbf{X}_{13}$ at $n = 7$, $\mathbf{X}_{23}$ at $n = 8$ and $\mathbf{X}_{33}$ at $n = 10$. With these blocks the fold is exact through λ^n at every n tested ($n = 4\text{--}9$, to 10^{-12}).

B. Two-sector ISR- $\text{ADC}(n)$ and the cross block

Two-sector ISR- $\text{ADC}(n)$ orthonormalizes and folds each sector on its own, and thus yields the familiar form of the self-energy, organized by + and – branches,

$$\Sigma(\omega) = \Sigma(\infty) + \mathbf{U}^{\text{II}}(\omega - \mathbf{K}^{\text{II}} - \mathbf{C}^{\text{II}})^{-1} \mathbf{U}^{\text{II}\dagger} + \mathbf{U}^{\text{I}}(\omega + \mathbf{K}^{\text{I}} + \mathbf{C}^{\text{I}})^{-1} \mathbf{U}^{\text{I}\dagger}. \quad (22)$$

Its blocks are exactly the same-sector blocks of $\tilde{\mathbb{M}}$, so it is $\tilde{\mathbb{M}}$ with $\mathbf{X}$ deleted. The cross block vanishes at the two lowest orders,

$$\mathbf{X}^{(0)} = \mathbf{X}^{(1)} = 0, \quad (23)$$

but not at second, where it is a contraction of the first-order doubles with the two-electron integrals. Expanding the union

fold in $\mathbf{X}$, the leading term the two-sector construction misses is

$$\Delta\Sigma^{(4)} = \mathbf{U}^{\text{I}(1)}(\omega + \mathbf{K}_1^{\text{I}})^{-1} \mathbf{X}_{11}^{(2)}(\omega - \mathbf{K}_1^{\text{II}})^{-1} \mathbf{U}^{\text{II}(1)\dagger} + \text{h.c.}, \quad (24)$$

which matches the λ^4 difference of the two folds to 5×10^{-18} . Two-sector ISR- $\text{ADC}(n)$ is therefore exact only through λ^3 , for every n .

The defect is fundamental, its complete-manifold limit is not FCI, as can be seen clearly in Fig. 1. This obstacle has been identified by Tarantelli and Cederbaum.⁵ They write the Dyson equation as a configuration-interaction problem in $\mathcal{H}_{N+1} \oplus \mathcal{H}_{N-1}$, with a composite Hamiltonian that is $\mathbb{M}$ of Eq. (16). The block that couples the $(N+1)$ and $(N-1)$ configurations vanishes in zeroth and first order but not in second, and they conclude that this coupling “prevents the possibility of reproducing analytically the fourth-order ADC scheme”, since ADC assumes that the self-energy splits into independent $(N+1)$ and $(N-1)$ parts. They remove the coupling by a further unitary transformation that block-diagonalizes the configuration space. This transformation can only be built order by order, and with an exactly solvable two-electron example they show that no transformation built from the ground state alone achieves the decoupling. The two-sector ISR is built from the ground state alone and is decoupled by construction.

C. Diagrammatic $\text{ADC}(n)$

Diagrammatic $\text{ADC}(n)$ removes $\mathbf{X}$ by the Tarantelli–Cederbaum rotation⁵ $\mathbf{W} = e^{\mathbf{A}}$, with $\mathbf{A}$ antisymmetric in the I–II blocks and

$$A_{\mu\nu}^{(m)} = \frac{R_{\mu\nu}^{(m)}}{K_\nu - K_\mu}, \quad (25)$$

$\mathbf{R}^{(m)}$ being the I–II part of $[\mathbf{W}^\dagger \tilde{\mathbb{M}} \mathbf{W}]^{(m)}$ from $\mathbf{A}^{(<m)}$; Eq. (20) is applied to the rotated blocks. At third order this moves $\mathbf{X}^{(2)}$ into the couplings,

$$\delta U_{pv}^{\text{II}(3)} = \sum_{\mu \in \text{I}_1} \frac{U_{p\mu}^{\text{I}(1)} X_{\mu\nu}^{(2)}}{K_\nu - K_\mu}, \quad (26)$$

where $K_\nu - K_\mu$ is a 3p3h energy; these rows reproduce the diagrammatic $\text{ADC}(4)$ rows⁷ to 10^{-18} .

D. Non-Dyson $\text{ADC}(n)$

Non-Dyson $\text{ADC}(n)^{8,9}$ builds the ISR from $\hat{r}_v^- \Psi_0$ in $\mathcal{H}_{N-1}$ (or $\hat{r}_v^+ \Psi_0$ in $\mathcal{H}_{N+1}$) with the ordinary overlap and the same rule; its eigenvalues are the IPs (EAs).

E. Perturbative references

$\Delta\text{MP}n$ is the RS series of $E_{N-1}(\lambda) - E_0(\lambda)$ for the Koopmans-connected state, summed through order n . $\text{MBGF}(n)^{10}$ takes

the exact $\Sigma^{[n]} = \sum_{m \leq n} \Sigma^{(m)}$ and solves

$$\omega = e_p(\omega), \quad (27)$$

with e_p the eigenvalue of $\epsilon + \Sigma^{[n]}(\omega)$ of largest weight on p , taking the root of largest residue. The pole Taylor series of union, diagrammatic and non-Dyson ADC(n) equal the exact one through λ^n to 10^{-17} ($n \leq 15$); the two-sector pole deviates from λ^4 on.

¹J. N. Lyness and C. B. Moler, *SIAM J. Numer. Anal.* **4**, 202 (1967).

²J. Schirmer, *Phys. Rev. A* **43**, 4647 (1991).

³F. Mertins and J. Schirmer, *Phys. Rev. A* **53**, 2140 (1996).

⁴J. Schirmer, *Many-Body Methods for Atoms, Molecules and Clusters* (Springer, Cham, 2018).

⁵A. Tarantelli and L. S. Cederbaum, *Phys. Rev. A* **45**, 2790 (1992).

⁶J. Schirmer, *Phys. Rev. A* **26**, 2395 (1982).

⁷J. Schirmer, L. S. Cederbaum, and O. Walter, *Phys. Rev. A* **28**, 1237 (1983).

⁸J. Schirmer, A. B. Trofimov, and G. Stelter, *J. Chem. Phys.* **109**, 4734 (1998).

⁹A. B. Trofimov and J. Schirmer, *J. Chem. Phys.* **123**, 144115 (2005).

¹⁰S. Hirata, I. Grabowski, J. V. Ortiz, and R. J. Bartlett, *Phys. Rev. A* **109**, 052220 (2024).